Field Robotics Deployment and Maintenance Report

Kestrel-7 Autonomous Survey Platform - Operating Season 2031

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Abstract

This report documents the deployment, sensor calibration, power management, and maintenance of the Kestrel-7 autonomous survey platform during the 2031 operating season. The platform completed twenty-six weeks of continuous field service across three survey sites. All values are derived from the platform's own instrument logs and are provided for demonstration and review purposes.

The report is organised into seven sections: an executive overview, sensor calibration, power management, field deployment, the maintenance schedule, safety procedures, and a final performance summary. A component-level maintenance table is included in the maintenance section.

1. Executive Overview

The Kestrel-7 autonomous survey platform completed a twenty-six week deployment across three survey sites during the 2031 operating season. Platform availability reached 94.6 percent of scheduled operating hours, with a total of 412 completed survey runs and eighteen scheduled maintenance interventions.

The platform demonstrated stable behaviour in all assigned operating conditions. The principal operational findings are summarised below:

- Sensor drift remained within the calibrated budget for the full season.

- No unplanned power interruptions were recorded on the primary battery rail.

- All maintenance actions were completed within their scheduled windows.

- The mean time between maintenance interventions was 22.9 operating hours.

The remainder of this report describes the calibration, power, deployment, maintenance, safety, and performance characteristics that produced these results.

2. Sensor Calibration

Sensor calibration is performed before every deployment cycle. The Kestrel frame is calibrated against the reference rig at the start of each cycle. The reference rig provides a known geometric baseline against which frame alignment is verified.

The inertial measurement unit (IMU) has a drift budget of 0.4 degrees per hour. Any drift measured above this budget requires the gyro module to be replaced before the next deployment.

The lidar array is aligned with the horizon prism using a two-point procedure. The two-point procedure is repeated whenever the payload mast is removed or the horizon prism is replaced. Calibration is verified with the reference rig at an ambient temperature within five degrees Celsius of the deployment temperature.

Between calibrations the reference rig is stored in the instrument case. The instrument case is transported with the platform at all times.

3. Power Management

The platform draws its operating power from two independent supplies. The inertial measurement unit is powered from the primary battery rail, which supplies 24 volts nominal. The communications module is powered from the auxiliary solar bus, which supplies 21 volts nominal under full illumination.

The battery management system disconnects the primary battery rail below 19.5 volts to protect the cell stack. Under normal survey load the platform endures for 9.5 hours on a single charge of the primary battery rail.

Auxiliary solar output is logged continuously. If the auxiliary solar bus falls below 17.5 volts the platform reduces the communications duty cycle until the bus recovers.

4. Field Deployment

Field deployment follows a fixed sequence. After the platform is positioned at the survey origin, the survey team waits four minutes after landing before initiating the lidar sweep. This settling period allows the inertial measurement unit to reach thermal equilibrium.

The deployment sequence then proceeds as follows: the communications module is raised and linked, the primary battery rail is enabled, the horizon prism is uncovered, and the first lidar sweep is initiated. The reference rig is carried in the instrument case and is unpacked only if a mid-deployment recalibration is required.

Deployment is suspended when the wind speed exceeds the limit stated in the safety procedures. The survey team records the deployment start and end time for every survey run.

5. Maintenance Schedule

Maintenance actions are scheduled from the platform's own operating-hour counter. The lidar array requires recalibration every 250 operating hours. The inertial measurement unit and the horizon prism are serviced every 500 operating hours. The primary battery rail and the communications module are inspected every 1000 operating hours.

The reference rig is recalibrated against the horizon prism every 500 operating hours so that the reference baseline itself remains traceable. The component-level schedule is given in the table below.

Component	Calibration Interval	Operating Range	Failure Threshold	Recommended Action
Lidar Array LR-9	250 h	0-80 m	> 2.5 mm error	Recalibrate on reference rig
Inertial Measurement Unit	500 h	-20-55 C	Drift > 0.4 deg/h	Replace gyro module
Horizon Prism	500 h	0.5-4.0 km	Misalignment > 0.2 deg	Re-align two-point procedure
Primary Battery Rail	1000 h	19.5-26.0 V	Below 19.5 V	Replace cell stack
Auxiliary Solar Bus	750 h	18.0-24.0 V	Output < 17.5 V	Clean array, test regulator
Communications Module	1000 h	2.4-5.8 GHz	Packet loss > 3%	Re-seat antenna
Payload Mast	2000 h	+/-15 deg tilt	Play > 1.5 mm	Torque mast bolts
Thermal Enclosure	1500 h	-10-60 C	Fan speed < 2000 rpm	Replace enclosure fan

Maintenance is recorded against the operating-hour counter rather than the calendar so that idle periods do not defer required service.

6. Safety Procedures

The maximum permissible wind speed for deployment is 12 metres per second. Deployment is suspended immediately when the measured wind speed exceeds this limit. A two-person rule applies to all deployments: a single operator may not launch, recover, or service the platform.

The emergency stop for the primary battery rail is located on the control mast. Battery handling follows the cell-stack procedure: the primary battery rail is isolated before any enclosure panel is opened. The thermal enclosure is allowed to cool for ten minutes before service.

7. Performance Summary

Over the 2031 operating season the platform recorded a mean time between failures of 121 operating hours and an availability of 94.6 percent. Observed drift averaged 0.31 degrees per hour, which remains below the 0.4 degree per hour budget defined in the sensor calibration section.

All eighteen maintenance interventions were completed within their scheduled windows. No safety incident was recorded. The platform is cleared for continued operation in the next scheduled season.